Learning Objective 4.1: I can describe FSK modulation and implement an FSK transmitter and receiver in GNU Radio.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Transmitter settings.** A flowgraph samples at $f_s = 48$ kHz and sends FSK at 1200 baud with a deviation of $\Delta f = 600$ Hz, one bit per symbol.

   (a) Find the samples per symbol to enter in the Repeat block.

   (b) Find the sensitivity to enter in the Frequency Mod block.

   (c) Find the modulation index and estimate the bandwidth with Carson's rule.

   h =

   B =

   (d) A classmate builds the same transmitter from two Signal Sources at $+600$ and -600 Hz and a selector that switches between them at each bit. Explain why Frequency Mod gives a more compact spectrum.

2. **Receiver settings.** The receiver for the link of Question 1 follows the L8 flowgraph: Low Pass Filter, Quadrature Demod, Moving Average, Symbol Sync, and Binary Slicer.

   (a) Find the Quadrature Demod gain that puts the two tones at $+1$ and -1.

   (b) Find the low-pass filter cutoff that passes the Carson bandwidth.

   (c) A classmate leaves the Quadrature Demod gain at 1. Find the levels the two tones produce, and explain whether the slicer still decides correctly.

   (d) The classmate also sets the filter cutoff to 8 kHz. By how many dB does that lower the SNR at the demodulator?

3. **Noise.** The L8 link samples at 32 kHz, sends 1000 baud with $\Delta f = 1$ kHz, and filters to the 4 kHz Carson bandwidth. The signal has power 1, and the Channel Model's noise of voltage σ has power σ^2 spread over the full 32 kHz.

   (a) Find the SNR at the demodulator for $\sigma = 0.8$.

   (b) The text starts to show errors below about 9 dB. Find the largest noise voltage that keeps the SNR at 9 dB.

   (c) The link doubles its symbol rate to 2000 baud and keeps $\Delta f = 1$ kHz. Find the new Carson bandwidth and the largest noise voltage for 9 dB.

   B =

   σ =

   (d) Lesson 2 designs a link with margin above the SNR it needs. Explain, from what the L8 link does between 9 dB and 3 dB, why margin matters for FSK.

4. **An FSK remote on a waterfall.** An RTL-SDR waterfall shows a burst with two tracks, at 433.900 and 433.940 MHz. In the alternating preamble, the signal stays on each track for 0.417 ms.

   (a) Find the center frequency, the deviation, and the symbol rate.

   f_c =

   Δf =

   R_s =

   (b) The remote's crystal is accurate to 10 ppm. Find the largest frequency offset at 433.92 MHz.

   (c) An FSK receiver tolerates an offset of about half its deviation. Explain whether a 4.3 kHz offset breaks the remote's link, and whether it would break the L8 link if that link ran at 433.92 MHz.

   (d) With a 600 Hz offset the L8 demodulator outputs levels near $+1.6$ and -0.4. Name one change to the receiver that would put them back at ± 1.

Documentation: _____